

SIT720 MACHINE LEARNING

D TASK MINI PROJECT

Part 1: Problem Definition and Data Collection

This project develops a housing price prediction and decision-support system for a Sydney real estate agency, using machine learning to support — not replace — informed valuation decisions.

Honestly, I picked Mosman, Paddington, and Campbelltown because they're basically three completely different price brackets I'd realistically end up looking at depending on where I was in life. Mosman was the "dream but let's be real" suburb — harbourside, prestige, the kind of place you'd want if money wasn't really a factor. Paddington felt like the more realistic inner-city professional option — still expensive, but you're paying for location and character rather than sheer size. Campbelltown was the practical end — somewhere an actual first-home buyer could genuinely afford to get into the market. I also liked that they weren't just expensive vs cheap — they're different housing types too: Mosman is mostly established houses, Paddington is heritage terraces, and Campbelltown has a mix of newer builds and standard family homes.

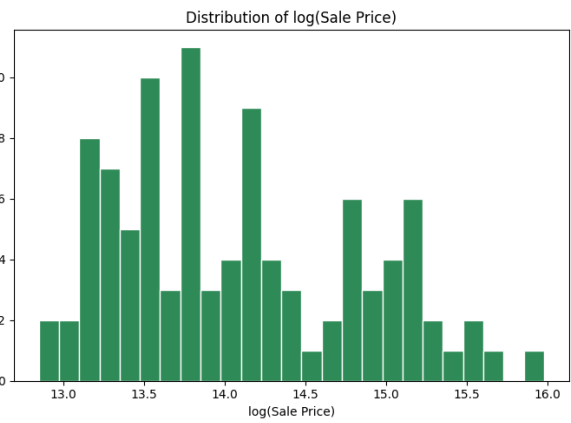
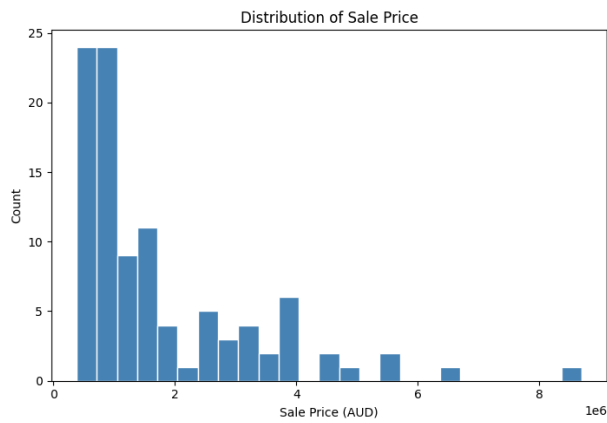
A dataset of 100 sold properties was manually collected from Domain.com.au (33 Mosman, 32 Paddington, 35 Campbelltown). Sale price was the prediction target, alongside bedrooms, bathrooms, car spaces, building size, year built, property type, school catchment, and the agent's listing description as text data. Prices ranged from \$380,000 to \$8.7 million, and median price-per-sqm varied nearly 4x between Paddington (\$23.7K/sqm) and Campbelltown (\$6.3K/sqm), confirming location as a dominant price driver.

Manual collection was time-consuming since every listing had to be read individually. Land size and distance-to-CBD/station weren't consistently available and were excluded, which likely limits the model's ability to capture accessibility effects. The dataset only captures sold properties (survivorship bias), and with 32–35 properties per suburb, the sample is still small relative to each suburb's actual housing diversity.

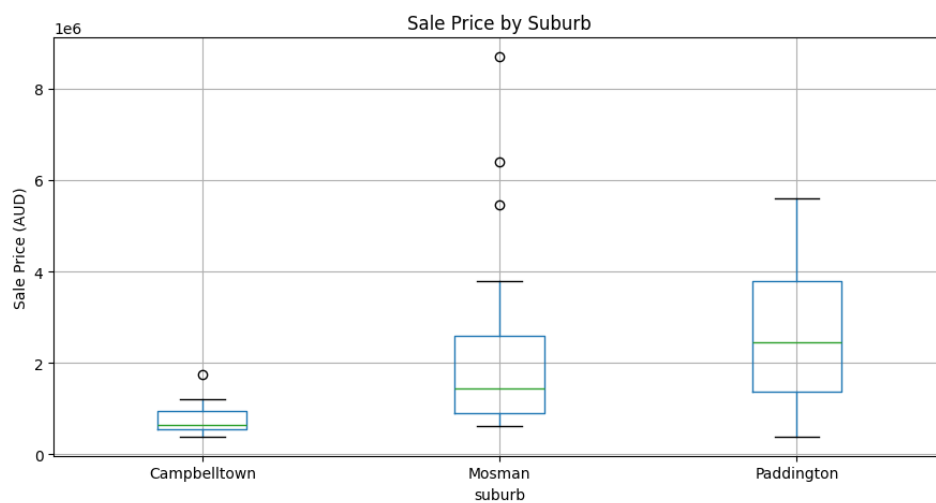
Part 2: Data Understanding and Feature Engineering

Sale prices range from \$380,000 to \$8.7 million, with a right-skewed distribution driven by a handful of high-value Mosman and Paddington sales. The three suburbs sit in clearly distinct price tiers — Campbelltown consistently cheapest and most tightly clustered, Mosman widest-spread due to a mix of standard homes and multi-million-dollar estates. Four statistical outliers were identified, all large Mosman/Paddington houses, and retained as genuine high-end market activity rather than errors.

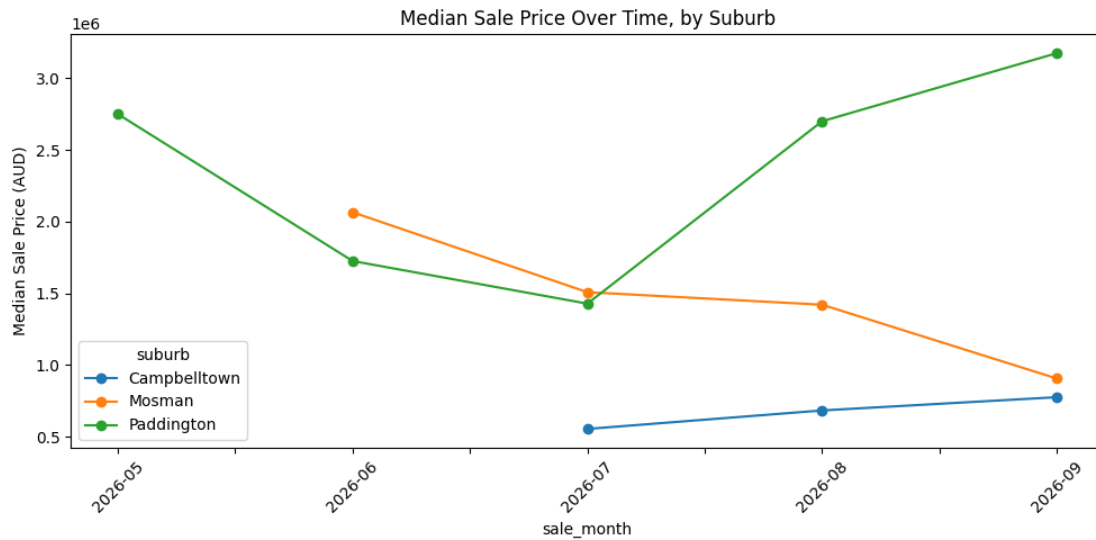
Before engineering any features, I expected suburb, bathrooms, and bedrooms to matter most — suburb because location clearly drives Sydney prices, and bathrooms/bedrooms as the clearest proxies for size and household suitability. This held up strongly: price-per-sqm is nearly 4x higher in Paddington (\$23.7K/sqm) than Campbelltown (\$6.3K/sqm), confirming location as the dominant driver. Where I was wrong was building size — it only correlated at 0.18 with price, far weaker than bathrooms or bedrooms, suggesting the number and type of rooms matters more here than raw floor area once suburb is accounted for.



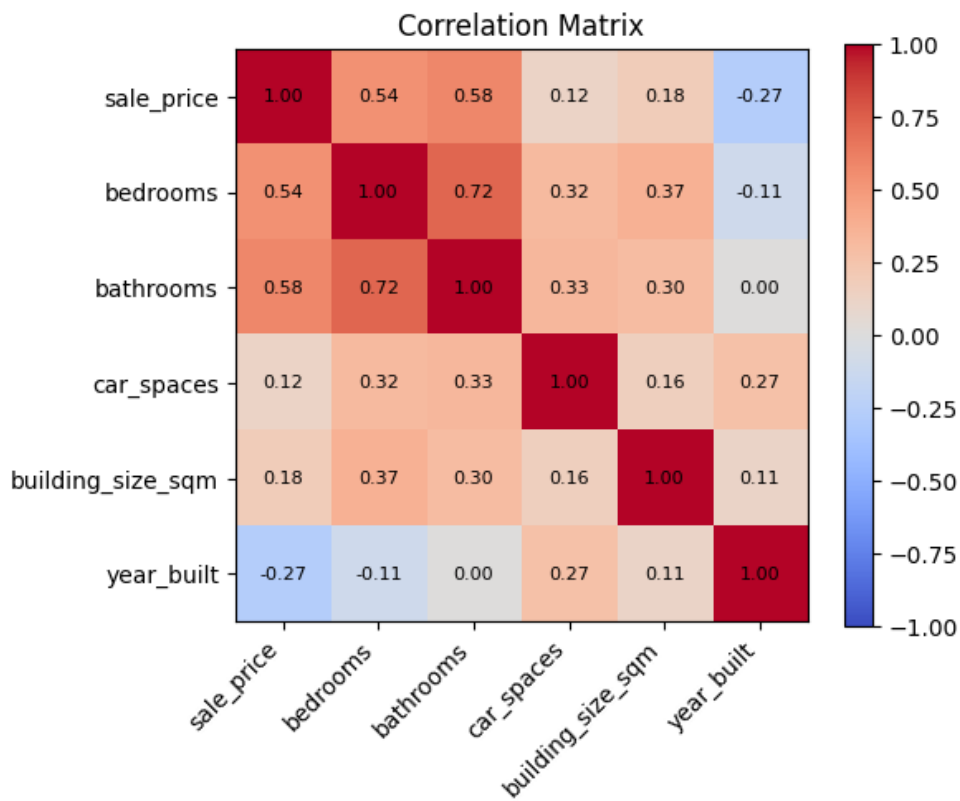
sale_price	
count	1.000000e+02
mean	1.751771e+06
std	1.516752e+06
min	3.800000e+05
25%	7.262500e+05
50%	1.155278e+06
75%	2.552000e+06
max	8.700000e+06



suburb						
Campbelltown	35	650000.0	7.631744e+05	2.815444e+05	386500	1750000
Mosman	33	1440000.0	2.079091e+06	1.826582e+06	610000	8700000
Paddington	32	2450000.0	2.495500e+06	1.423887e+06	380000	5600000



	suburb	address	sale_price	property_type	bedrooms	building_size_sqm
67	Mosman	5 Botanic Road	8700000	House	4	290
80	Mosman	5 Central Avenue	6399000	House	5	350
94	Paddington	76A Paddington Street	5600000	House	3	189
79	Mosman	17 Avenue Road	5450000	Apartment	5	255



	sale_price
sale_price	1.000000
bathrooms	0.584704
bedrooms	0.537980
building_size_sqm	0.181194
car_spaces	0.119830
year_built	-0.273122

	suburb	sale_price	property_age	price_per_sqm	total_rooms
0	Mosman	980000	25	13066.666667	2
1	Mosman	871000	59	14516.666667	3
2	Mosman	905000	22	15083.333333	3
3	Mosman	730000	66	18250.000000	2
4	Mosman	750000	60	18292.682927	2

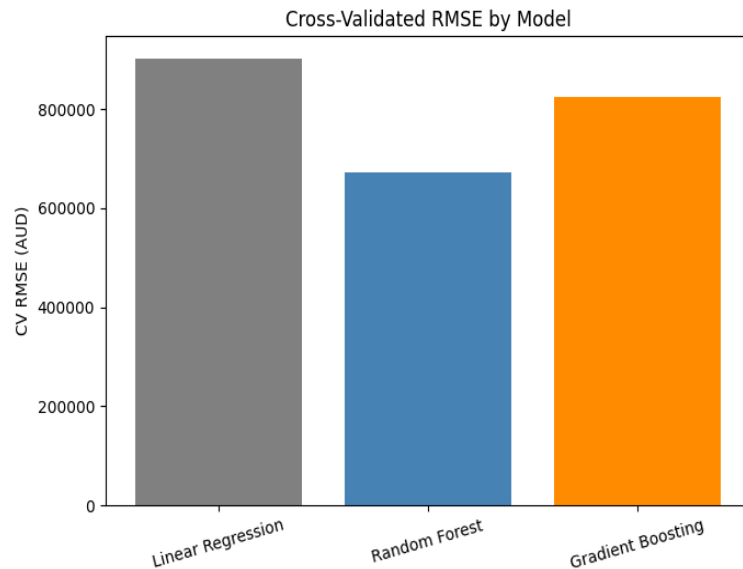
	price_per_sqm
suburb	
Paddington	23755.299105
Mosman	16666.666667
Campbelltown	6333.333333
dtype:	float64

Part 3: Model Development and Evaluation

Before training, I expected Random Forest or Gradient Boosting to beat Linear Regression, since suburb effects are categorical and interact with size in ways linear regression can't capture. I guessed Gradient Boosting would edge out Random Forest, though worried it might overfit given the small dataset (~100 rows).

Random Forest was chosen as the final model, with the best cross-validated performance overall — lowest CV RMSE (\$675K vs \$910K for Linear Regression and \$793K for Gradient Boosting), highest CV R^2 (0.745), and the smallest overfitting gap (0.21 vs 0.26 and 0.358 respectively). Linear Regression underperformed because it treats suburb as a flat adjustment rather than an interaction with size. Gradient Boosting, despite a near-perfect training R^2 (0.997), overfit badly — CV R^2 dropped to 0.639 — a known risk for high-capacity models on small datasets. Random Forest avoided this by averaging across decorrelated trees, and on the held-out test split achieved $R^2 = 0.947$, RMSE $\approx$ \$354K. My prediction was half right: tree-based models did beat Linear Regression, but Random Forest won because Gradient Boosting's extra flexibility became a liability on this dataset size.

	CV RMSE	CV MAE	CV R2	Train R2 (avg)	R2 gap (train - CV)
Linear Regression	901613.708	656894.447	0.520	0.773	0.253
Random Forest	673216.236	405761.491	0.743	0.955	0.213
Gradient Boosting	824284.973	481874.314	0.586	0.997	0.411



Part 4: Investigating Prediction Failures

Looking at the five properties with the largest prediction errors, a pattern jumps out — four of the five are in Mosman, and the model consistently struggles with precision on Mosman's most expensive properties, sometimes over-predicting (177A Spit Road: actual \$3.2M vs predicted \$3.9M) and sometimes under-predicting (5 Central Avenue: actual \$6.4M vs predicted \$5.75M), so it's not a one-directional bias.

The 10/16 Avenue Road Mosman apartment stands out the most — 55.8% error, far bigger than the others. It's a small 80sqm, 2-bed apartment that sold for only \$892,000, but the model predicted \$1.39M — likely because the model has learned "Mosman = expensive" from all the multi-million-dollar houses in the suburb, and doesn't have enough small-apartment examples there to correct for that. The Paddington terrace (28 Napier Street) is also worth mentioning — a large, 136-year-old property is a type the model has few examples of, so lower reliability here makes sense.

What's missing that could help: none of my features capture renovation status, interior condition, or view — all of which matter a lot for exactly these high-value, unique properties. Overall, predictions should be treated more cautiously for expensive, low-frequency property types within a suburb (Mosman apartments, older heritage terraces), since the model has fewer comparable examples to learn from compared to the more "typical" 3-4 bed houses that make up the bulk of the dataset.

	address	suburb	property_type	bedrooms	bathrooms	building_size_sqm	actual_price	predicted_price	abs_error	pct_error
22	28 Napier Street	Paddington	Terrace	4	2	177	450000	3.690969e+06	809030.620028	17.978458
44	177A Spit Road	Mosman	House	3	2	245	320000	3.903095e+06	703095.089263	21.971722
80	5 Central Avenue	Mosman	House	5	4	350	6399000	5.752286e+06	646713.833333	10.106483
70	10/16 Avenue Road	Mosman	Apartment	2	1	80	892000	1.389652e+06	497651.778446	55.790558
77	2/81A Glover Street	Mosman	Apartment	2	2	97	1440000	1.812474e+06	372474.496475	25.866284

Part 5: Human Judgement, Machine Learning, and Large Language Models

Ten held-out test properties were compared across three approaches: the trained Random Forest model, an LLM prompted with each property's features, and my own estimate based on market knowledge from this project.

The LLM performed best overall, with a mean absolute error of \$413,700 (15.8% average error), narrowly ahead of my own human estimates at \$436,700 (16.3%), with the ML model showing a lower raw dollar error (\$375,468) but the highest percentage error (21.9%) due to struggling more at the extremes of the price range.

Breaking it down property by property, the ML model won on both Paddington properties and the largest Mosman house, suggesting it captured suburb-level patterns well for properties closer to a suburb's typical profile. My own estimates and the LLM's split the remaining Mosman cases fairly evenly — human judgement edged ahead on properties that felt like outliers for their suburb (a small under-priced apartment, an older house), where reasoning "this seems atypical" helped, while the LLM did better on more standard mid-range properties.

This suggests human judgement still adds value particularly for atypical properties that don't fit a suburb's usual pattern, something neither the ML model nor the LLM — both working from limited structured features — fully capture. At the same time, the LLM's strong performance without ever seeing this specific dataset shows general market knowledge can rival a purpose-built model trained on a small sample.

ML model (mean abs error)	375467.781677
LLM (mean abs error)	413700.000000
Human (mean abs error)	436700.000000
dtype: float64	

	address	suburb	actual_price	ml_prediction	llm_prediction	human_prediction
45	5/32 Broughton St	Campbelltown	620000	7.711178e+05	650000	650000
76	2/60A Raglan Street	Mosman	1175000	1.437275e+06	1050000	1000000
22	28 Napier Street	Paddington	4500000	3.690969e+06	3600000	3500000
18	54 Stewart Street	Paddington	3965000	3.867945e+06	3400000	3300000
70	10/16 Avenue Road	Mosman	892000	1.389652e+06	1200000	1150000
90	40/31-35 Chamberlain Street	Campbelltown	515000	6.353433e+05	580000	600000
44	177A Spit Road	Mosman	3200000	3.903095e+06	3750000	3700000
77	2/81A Glover Street	Mosman	1440000	1.812474e+06	1700000	1650000
80	5 Central Avenue	Mosman	6399000	5.752286e+06	5100000	5000000
53	4/2A Condamine Street	Campbelltown	635000	7.299204e+05	600000	590000

Part 6: Final Deployment and Reflection

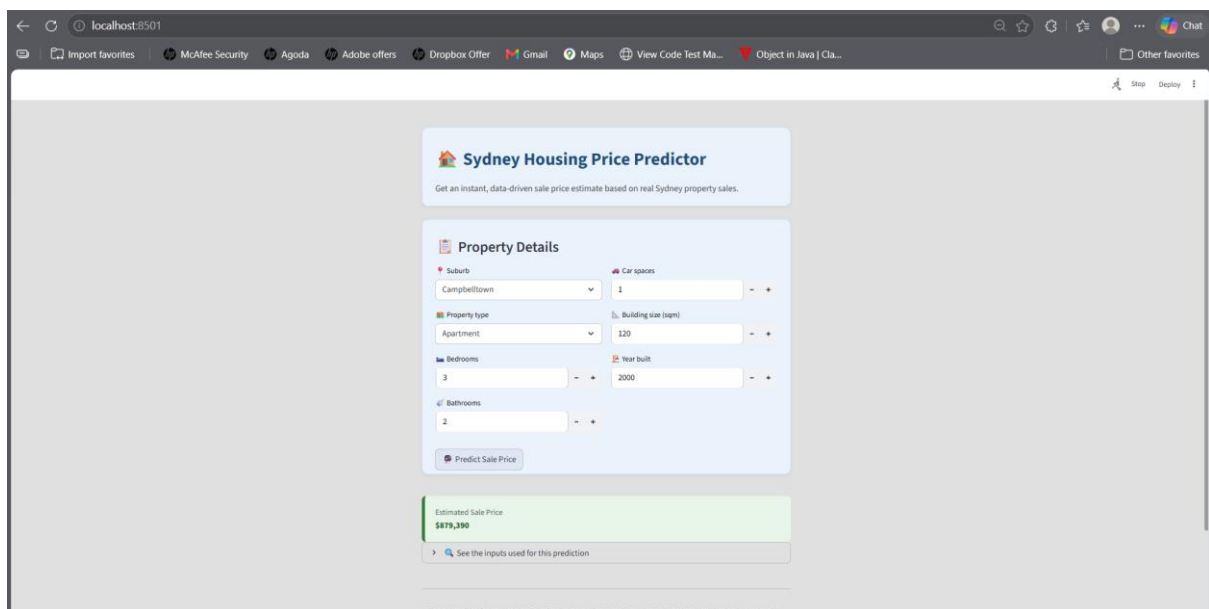
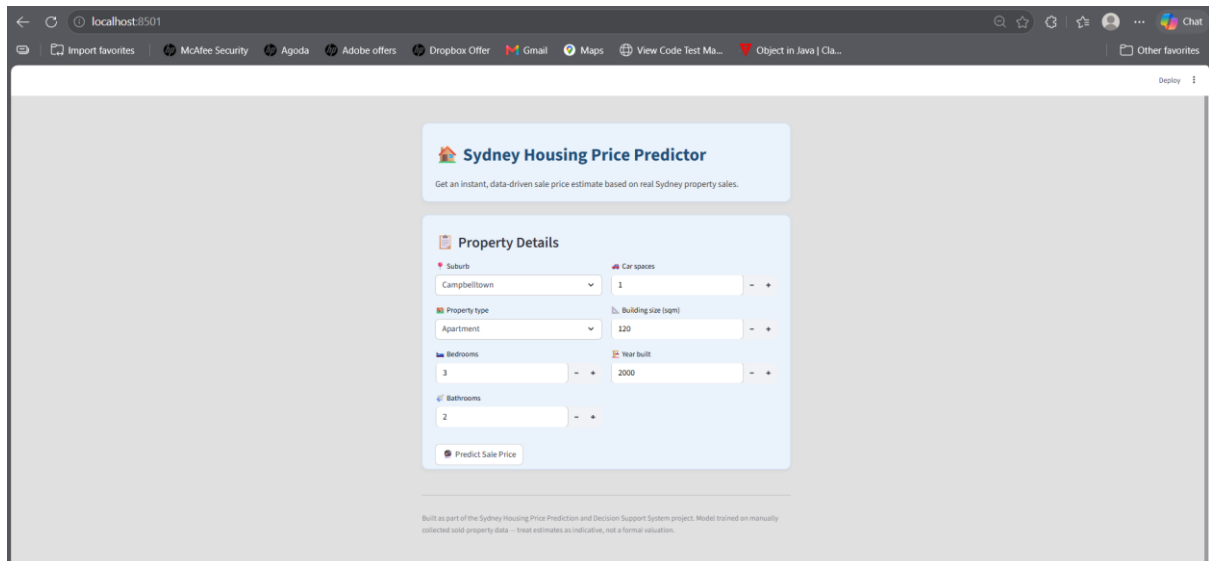
Application development

A Streamlit web application was developed to serve the final trained model. The user enters suburb, property type, bedrooms, bathrooms, car spaces, building size, and year built, and receives a predicted sale price. The app loads the fitted scikit-learn pipeline (including one-hot encoding of categorical

features) via joblib, and displays results in a styled result card alongside an expandable view of the exact inputs used.

Build and run instructions

1. Install dependencies: `pip install streamlit scikit-learn joblib pandas`
2. Ensure `sydney_housing_model.pkl` and `model_metadata.pkl` are in the same folder as `app.py`
3. Run: `streamlit run app.py`
4. The app opens automatically at <http://localhost:8501>



Critical reflection

The biggest challenge wasn't modelling — it was data collection. Manually reading and transcribing Domain listings across three suburbs took far longer than expected, and some features (land size for units, transport distance) weren't consistently available and had to be excluded, likely limiting the model's ability to capture accessibility effects.

The clearest modelling lesson was watching overfitting happen in real time: Gradient Boosting's near-perfect training score didn't hold up on cross-validation, while Random Forest generalised better despite less impressive training metrics — a reminder that predictive performance and reliability aren't the same thing, and the model I deployed prioritises the latter.

Ethically, the main concern is sample imbalance across suburbs — only 32–35 properties each, which Part 4 showed directly causes over-prediction on Mosman's less common property types. In a higher-stakes context (lending, insurance) this kind of imbalance could produce genuinely unfair outcomes. With more time, I'd collect a larger, more balanced dataset per suburb, add land size and transport-distance features, and gather data over a longer period to capture real price trends rather than a few months' snapshot.

Reference:

[“https://www.domain.com.au/?gad_source=1&gad_campaignid=875283278&gbraid=0AAAAAD9U5_Kvbzsv7xXPocIWcfIH1r-x2&gclid=Cj0KCQjwt9jVBhDXARIsAFSP-6d_CMMNP1fCRuVILWCiJdplpNZyBV_Xni7y5uoISpr8qqicroYPYasaAlZgEALw_wcB”](https://www.domain.com.au/?gad_source=1&gad_campaignid=875283278&gbraid=0AAAAAD9U5_Kvbzsv7xXPocIWcfIH1r-x2&gclid=Cj0KCQjwt9jVBhDXARIsAFSP-6d_CMMNP1fCRuVILWCiJdplpNZyBV_Xni7y5uoISpr8qqicroYPYasaAlZgEALw_wcB)

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